Columnar Transposition Cipher

Aim

To implement the **Columnar Transposition Cipher encryption technique** in C by arranging the plaintext in rows under a keyword and reading the columns in the alphabetical order of the keyword to generate the ciphertext.

Algorithm

1. **Start** the program.
2. Enter the plaintext message.
3. Enter the keyword.
4. Determine the number of columns from the length of the keyword.
5. Arrange the plaintext row by row in a matrix with columns equal to the keyword length.
6. If the last row is incomplete, fill the remaining cells with **'X'**.
7. Number the columns according to the alphabetical order of the keyword.
8. Read the characters column by column in the sorted keyword order.
9. Store the result as the ciphertext.
10. Display the ciphertext.
11. **Stop** the program.

PROGRAM :

```
#include <stdio.h>
#include <string.h>

int main()
{
    char plaintext[100], keyword[20];
    char matrix[20][20];
    int order[20];
    int i, j, k, rows, cols, len;

    printf("Enter Plaintext: ");
    scanf("%s", plaintext);

    printf("Enter Keyword: ");
```

```
scanf("%s", keyword);

len = strlen(plaintext);
cols = strlen(keyword);
rows = (len + cols - 1) / cols;

// Fill matrix row-wise
k = 0;
for(i = 0; i < rows; i++)
{
    for(j = 0; j < cols; j++)
    {
        if(k < len)
            matrix[i][j] = plaintext[k++];
        else
            matrix[i][j] = 'X';
    }
}

// Determine column order
for(i = 0; i < cols; i++)
    order[i] = i;

for(i = 0; i < cols - 1; i++)
{
    for(j = i + 1; j < cols; j++)
    {
        if(keyword[order[i]] > keyword[order[j]])
        {
            int temp = order[i];
            order[i] = order[j];
            order[j] = temp;
```

```
            }
        }
    }

    printf("\nCiphertext: ");

    // Read columns according to sorted keyword
    for(i = 0; i < cols; i++)
    {
        int col = order[i];
        for(j = 0; j < rows; j++)
        {
            printf("%c", matrix[j][col]);
        }
    }

    return 0;
}
```

OUTPUT :